NEUROMORPHIC COMPUTING — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on Neuromorphic Computing

Question 1

10.0 points

Examine the diagram below of a leaky integrate-and-fire (LIF) neuron model, which shows the membrane potential trace over time. Analyze how the membrane potential changes in response to incoming spikes and explain the conditions under which the neuron fires. Specify any key parameters illustrated in the diagram, such as threshold potential and leakage rate.

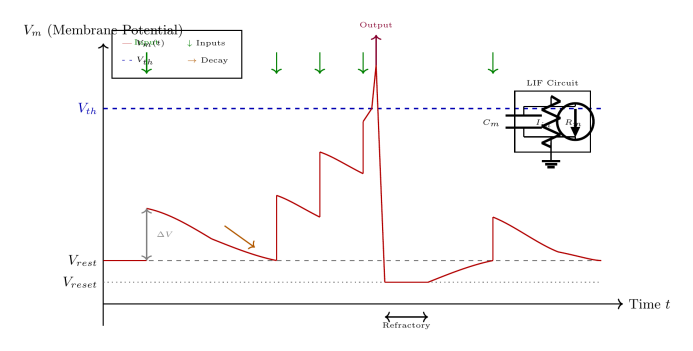

Fig. 1. Diagram for Question 1

Question 2

10.0 points

Examine the diagram below illustrating a spike-timing dependent plasticity (STDP) learning rule. Describe how the timing of pre- and postsynaptic spikes affects the synaptic weight change. Discuss the implications of synaptic modifications based on different time intervals between spikes.

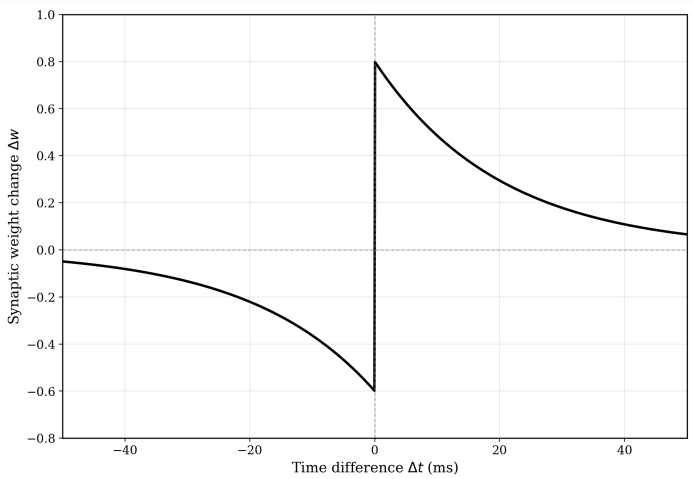

Fig. 2. Diagram for Question 2

Question 3

10.0 points

Examine the diagram below of a crossbar array architecture used for storing synaptic weights in neuromorphic computing. Explain how this architecture enables efficient weight storage and retrieval. Highlight any challenges related to scaling and variability illustrated in the diagram.

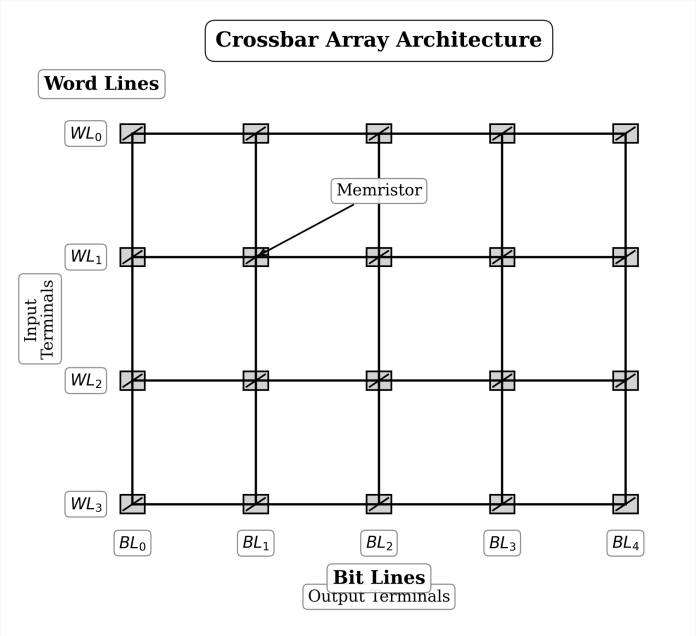

Fig. 3. Diagram for Question 3

Question 4

10.0 points

Evaluate the temporal coding scheme in the spiking neural network diagram below. Describe how information is encoded temporally and its impact on network processing capabilities. What are the advantages of temporal coding compared to rate coding based on the diagram?

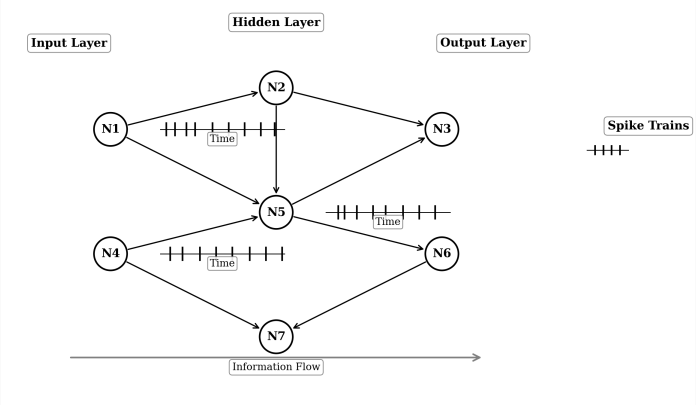

Fig. 4. Diagram for Question 4

Question 5

10.0 points

Examine the comparative diagram of energy consumption between von Neumann and neuromorphic architectures shown below. Discuss the reasons for differences in energy efficiency. What implications does this have for large-scale neural applications?

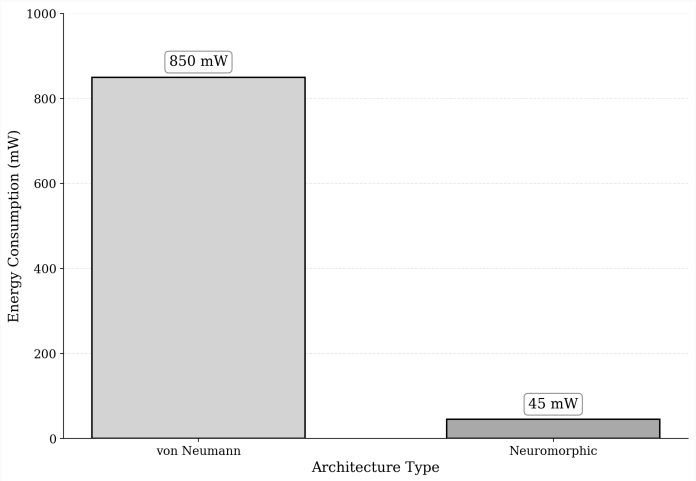

Fig. 5. Diagram for Question 5